

QUANTIZER GEOMETRY MATTERS: TASK-AWARE KV-CACHE COMPRESSION FOR LANGUAGE MODEL INFERENCE

Anonymous authors

Paper under double-blind review

ABSTRACT

KV-cache compression reduces the memory and bandwidth cost of language-model inference, but its effects depend on how the cache is used. We study ordinary language-model quality, long-context retrieval, and speculative decoding, where a compressed draft cache affects token acceptance while an unquantized target still preserves final correctness. Our initial Gaussian-perturbation experiments suggested that keys intrinsically require more precision than values. Controlled quantization rejects that interpretation: conclusions about key versus value precision depend critically on quantizer geometry. A naive per-token symmetric quantizer makes 8-bit keys with 4-bit values (K8V4) outperform equal-memory K4V8 by 10.87 acceptance points across six model pairs. Replacing only the key quantizer with grouped per-channel affine quantization changes the same contrast to -0.25 points; the paired geometry shift is -11.12 points (95% model-pair bootstrap CI $[-19.71, -4.25]$). This reveals that persistent key-channel outliers—not an intrinsic need for more key bits—drive the apparent asymmetry. This motivates a matched study of mixed-precision allocations using KIVI-style per-channel affine keys and per-token affine values under ordinary language-model quality, retrieval, and speculative acceptance. Across six speculative model pairs, K4V4 removes 66.13% of the draft cache and 22.64% of total target-plus-draft KV at 1K, with a -0.79 point acceptance change (95% CI $[-1.63, +0.04]$), ordinary-quality KL 0.00730, and 95.99% top-1 agreement. In ten strict exact-byte matrices, acceptance-aware allocation selects the same policies as quality-aware allocation at mild budgets and is significantly worse at the aggressive budget. Thus, acceptance remains an evaluation metric, but our acceptance-specific additive objective provides no allocation benefit. On 192 associative-retrieval examples, changing only the key quantization axis changes the equal-memory K4V2–K2V4 accuracy gap from $+81.25$ to $+16.15$ points (interaction -65.10 points, 95% CI $[-73.44, -56.77]$), showing that precision allocation and quantizer geometry cannot be selected independently. An actual bit-packed implementation confirms that K4V4 reduces a Qwen2.5-1.5B-shaped 32K cache from 896.0 to 260.2 MiB (70.96%). Across six draft-model shapes, three batch sizes, and five contexts from 1K to 32K, a direct packed Triton microkernel is geometrically $3.43\text{--}3.90\times$ faster than materialize-then-attend, depending on precision, but no tested cell beats native BF16 SDPA. This is a storage-feasibility result, not a production-throughput claim. The target verifier is never quantized; we separately audit finite-precision differences between batched verification and tokenwise BF16 decoding.

1 INTRODUCTION

Speculative decoding converts several serial target-model decoding steps into a single parallel verification step by sampling proposals from a cheaper draft model (Leviathan et al., 2023; Chen et al., 2023). The method is distribution preserving, but its systems footprint includes two model states and two growing KV caches. This duplication becomes consequential at long context, where each generated token reads the prefix cache and cache capacity limits the number of concurrently served requests.

Quantizing the draft cache seems straightforward, but generic model-quality metrics do not reveal whether a compressed cache remains a useful speculator. A small distributional error can move the draft’s top proposal away from the target’s token and terminate an entire speculative block. Conversely, errors in directions irrelevant to the verifier may have little effect on acceptance. This motivates measuring compression inside the speculative loop and testing whether an acceptance-aware objective improves upon ordinary language-model quality.

Our initial hypothesis was stronger: because keys determine attention routing, we expected keys to be intrinsically less compressible than values and expected speculative acceptance to require a distinct precision-allocation objective. Both ideas are plausible but incomplete. Isotropic Gaussian noise makes keys look dramatically more fragile, yet noise is not a cache quantizer and ignores the persistent channel-wise outliers that determine code utilization. Once we control quantization axis, grouping, zero-points, and actual bytes, the apparent K-first rule weakens or reverses. Likewise, acceptance-aware calibration does not outperform ordinary-quality calibration in our strict held-out matrices. This negative result changes the paper’s focus from proposing a special speculative objective to identifying which experimental factors genuinely determine KV-compression behavior.

Our investigation starts from the attention decomposition

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M\right), \quad O_\ell = A_\ell V_\ell W_{O,\ell}. \quad (1)$$

Keys determine the routing distribution A_ℓ , whereas values determine the payload read after routing. We began by hypothesizing that coarse key quantization would be more likely to redirect attention and cascade through subsequent layers than an equal-memory reduction in value precision. This role asymmetry is compatible with prior KV quantization analyses (Liu et al., 2024; Hariri et al., 2025), but we ask a different operational question: *which errors can a draft model tolerate without losing speculative acceptance?*

Our investigation separates three questions that prior drafts conflated. First, Gaussian perturbation establishes local sensitivity but does not specify a good quantizer. Second, quantization axis, grouping, and zero-points determine whether key outliers consume the available code range. Third, after controlling the quantizer, speculative acceptance and ordinary language-model quality may rank equal-memory allocations differently. We evaluate each question independently instead of interpreting reconstruction or perturbation error as a precision allocation rule. The first two questions produce a strong geometry effect; the third produces a useful null result rather than a new objective-specific method.

2 BACKGROUND AND PROBLEM FORMULATION

Let $p(\cdot \mid x_{\leq t})$ be the target distribution and $q(\cdot \mid x_{\leq t}; \hat{\mathcal{C}}_d)$ the draft distribution produced from a compressed draft cache $\hat{\mathcal{C}}_d$. Standard speculative sampling accepts proposals using the target correction rule; greedy speculative decoding accepts the longest proposal prefix matching target argmax tokens. In both cases, target verification is distribution preserving in exact arithmetic: compression affects efficiency through acceptance, not the target correction rule. In finite-precision greedy experiments, batched and token-wise target evaluation can choose different tokens when the largest BF16 logits are tied. We therefore report both sequence agreement and the target top-1 margin at every disagreement, distinguishing numerical ties from larger kernel-path differences.

For target and draft cache sizes M_t and M_d , native speculative decoding uses $M_t + M_d$. If layer ℓ stores keys and values with precisions b_ℓ^K, b_ℓ^V , the compressed draft cost is approximately

$$\widehat{M}_d \propto T \sum_{\ell=1}^{L_d} H_\ell^{KV} d_{h,\ell} (b_\ell^K + b_\ell^V), \quad (2)$$

plus quantizer metadata and any full-precision residual window. Our simulator counts the packed payload and either one 16-bit scale for a symmetric group or a 16-bit scale and offset for an affine group. Total saving is $(M_d - \widehat{M}_d)/(M_t + M_d)$; it is necessarily smaller than the percentage saved from the draft cache alone.

3 METHOD

3.1 DRAFT-CACHE FAKE QUANTIZATION

We distinguish a deliberately naive diagnostic quantizer from the outlier-aware quantizer used in our main matched experiments. The diagnostic baseline applies symmetric per-token fake quantization to either a key or value vector x ,

$$s(x, b) = \frac{\max_i |x_i|}{2^{b-1} - 1}, \quad \hat{x} = s \text{ clip}(\text{round}(x/s), -2^{b-1} + 1, 2^{b-1} - 1). \quad (3)$$

The key-axis ablation changes only keys to grouped per-channel affine quantization over 32 consecutive tokens and retains symmetric per-token values. The main matched-objective quantizer follows the full KIVI geometry: keys use the same per-channel groups, while values use affine groups over each token-KV-head vector. For every affine group G , we compute

$$s_G = \frac{\max G - \min G}{2^b - 1}, \quad z_G = \text{round}(-\min G/s_G), \quad (4)$$

and reconstruct with

$$\hat{x} = s_G [\text{clip}(\text{round}(x/s_G) + z_G, 0, 2^b - 1) - z_G]. \quad (5)$$

We test both no residual window and a 128-token BF16 residual window; in the latter, only older cache entries are quantized. Only the draft KV state is quantized between cached decode steps. The target model and verifier remain in BF16. This simulator measures the algorithmic memory-acceptance tradeoff without conflating it with a particular packed-cache kernel.

The evaluator performs one target and one draft prefill per prompt. Each draft round starts from the retained draft prefix cache, while the target verifies the proposal in one cached forward pass. Accepted states are committed and rejected suffix states are cropped in place; neither model recomputes the long prefix. Target-greedy references are generated once outside the timed configuration loop. This cache-resident execution is essential at long context and prevents prefix recomputation from dominating either memory or runtime measurements.

3.2 ACCEPTANCE-AWARE SENSITIVITY PROFILES

We first evaluate a full-precision draft baseline. Then, for each selected layer ℓ , component $c \in \{K, V\}$, and candidate bit width $b \in \{4, 8\}$, we quantize only (ℓ, c) while leaving all other draft-cache entries in BF16. The measured risk is

$$\hat{r}_{\ell,c,b}^{\text{SD}} = \overline{\alpha_{\text{BF16}} - \alpha_{\ell,c,b}}, \quad (6)$$

where α is the end-to-end speculative acceptance rate and the bar denotes a paired mean over calibration prompts. Because a small profile can make harmful quantization appear beneficial by chance, we also evaluate the one-sided uncertainty-aware risk

$$r_{\ell,c,b}^{\text{SD,UCB}} = \max(0, \hat{r}_{\ell,c,b}^{\text{SD}} + 1.96 \text{SE}(\alpha_{\text{BF16}} - \alpha_{\ell,c,b})). \quad (7)$$

As a smoother alternative to discrete greedy accept/reject events, we profile the speculative acceptance mass

$$A(p, q) = \sum_y \min\{p(y), q(y)\} = 1 - \text{TV}(p, q), \quad (8)$$

which is the expected one-step acceptance probability under the standard stochastic speculative-decoding correction rule. We evaluate allocations from both the paired acceptance-rate UCB and the paired acceptance-mass UCB; neither surrogate is selected using held-out evaluation outcomes. We additionally log accepted tokens per verification round, top-1 match, acceptance mass, Jensen-Shannon divergence, greedy target agreement, and the target top-1 margin at the first disagreement.

3.3 ORDINARY QUALITY VERSUS SPECULATIVE ACCEPTANCE

The same frozen draft model is profiled under a second downstream objective: teacher-forced continuation quality without speculative decoding. For each candidate we measure

$$r_{\ell,c,b}^{\text{LM}} = \mathbb{E}_t[D_{\text{KL}}(p_{\text{BF16}}(\cdot | x_{\leq t}) \parallel p_{\ell,c,b}(\cdot | x_{\leq t}))]. \quad (9)$$

This paired design isolates the downstream objective: model, prompts, quantizer, candidate bit widths, and memory budget are held fixed, while the allocation is optimized either for ordinary LM quality or speculative acceptance. Each selected allocation is then cross-evaluated under both objectives.

Table 1: Cross-family evaluation. Each pair is tested at 1K and 4K context, followed by top-eight-layer K/V profiling and held-out allocation evaluation.

Family	Target	Draft
Qwen2.5	3B, 7B	1.5B, 3B
Qwen3	8B	4B
Llama 3	Llama-3.1-8B	Llama-3.2-3B
OLMo 2	7B	1B
SmolLM2	1.7B	360M

3.4 EQUAL-MEMORY MIXED-PRECISION ALLOCATION

Given one-at-a-time risks r^o for objective o , our transparent allocator uses exact dynamic programming to solve the additive surrogate

$$\min_{\{b_\ell^c\}} \sum_{\ell, c} r_{\ell, c, b_\ell^c}^o \quad \text{s.t.} \quad \frac{1}{2|\mathcal{L}|} \sum_{\ell \in \mathcal{L}, c} b_\ell^c = B, \quad b_\ell^c \in \{4, 8, 16\}. \quad (10)$$

Thus quality- and acceptance-optimized allocations consume exactly the same profiled-layer bit budget B . We additionally compare against matched-memory K-priority and V-priority heuristics (e.g., K8V4 and K4V8 for $B = 6$). The additive approximation is validated by running every allocation through the complete cached decoding loops; it is not treated as the final performance estimate.

4 EXPERIMENTAL SETUP

We evaluate the target–draft pairs in Table 1 using WikiText-2 validation blocks, four draft proposals per verification round, greedy target verification, and 16 generated tokens per prompt. The objective matrix uses 32 held-out sequences, three shuffle seeds, contexts 512, 1K, and 4K, and profiled mean budgets $B \in \{6, 8, 10, 12\}$. We separately repeat the Qwen matrix on streaming C4 and compare 16-versus 64-prompt acceptance calibration. Main profiles cover the top eight draft layers; an all-layer Qwen run measures the larger attainable memory reduction. Domain-shift experiments use GSM8K answers and HumanEval prompts, while PG19 supplies 16K–32K prefixes. Cross-family evaluation includes Qwen2.5, Qwen3, Llama 3, OLMo 2, and SmolLM2. We also vary the proposal length over $\gamma \in \{2, 4, 8\}$ to test whether allocation rankings change as rejection wastes longer draft blocks. All pairs are checked for tokenizer compatibility.

We report acceptance rate, accepted tokens per target verification, ordinary-LM KL, continuation NLL, top-1 match, JS divergence, acceptance mass, draft-cache bytes, and total target-plus-draft KV bytes. Confidence intervals use paired prompt bootstrap resampling. Acceptance is estimated as the ratio of total accepted to total proposed tokens within each bootstrap sample, rather than an unweighted mean of per-prompt rates. Exactness audits separate target disagreements with top-1 margin at most 10^{-3} from non-tie mismatches and exclude the latter from paired acceptance inference. Python fake quantization does not provide a valid kernel-level speed measurement, so we do not make throughput claims from the quality evaluator. We separately validate persistent storage using actual bit-packed payloads and report its unfused materialize-then-attend latency only as a diagnostic.

5 RESULTS

5.1 ADDITIVE NOISE MAKES KEYS APPEAR FRAGILE

Figure 1 perturbs all draft-cache keys or values with Gaussian noise scaled to the clean activation standard deviation. At $\alpha = 0.03$, key noise reduces top-1 agreement with the clean draft to 0.617 and raises JS divergence to 0.176. Value noise at the same magnitude retains 0.992 top-1 match with JS divergence 0.0004. At $\alpha = 0.1$, the corresponding top-1 matches are 0.109 for K and 0.953 for V. This large separation is consistent with routing errors in K changing A in Equation 1 before values are read.

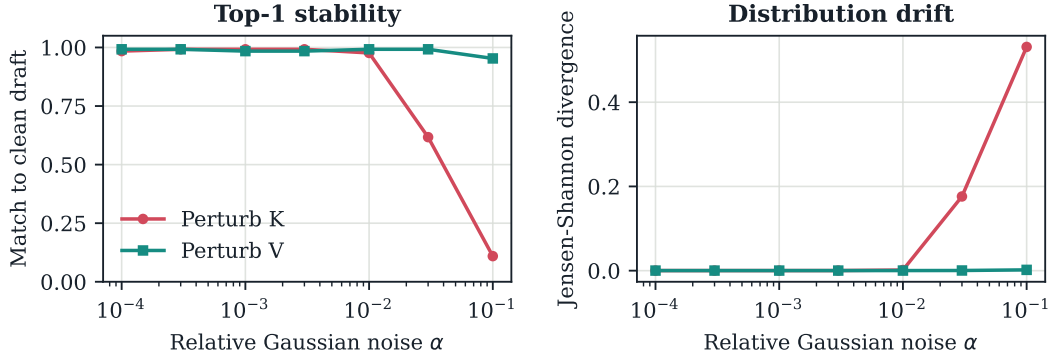


Figure 1: Controlled perturbations of the Qwen2.5-1.5B draft cache. Keys are far more sensitive to isotropic Gaussian noise, but this experiment does not model an outlier-aware quantizer.

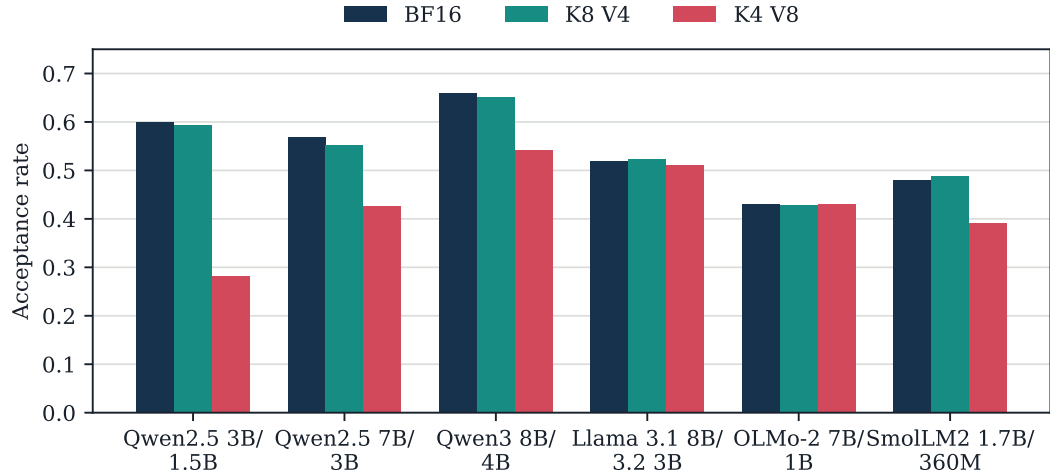


Figure 2: Acceptance at 1K WikiText context under the naive per-token symmetric quantizer. This comparison is a diagnostic of quantizer geometry, not evidence that keys intrinsically require more bits.

5.2 NAIVE PER-TOKEN QUANTIZATION CREATES AN APPARENT ASYMMETRY

Figure 2 isolates the core comparison at 1K context under per-token symmetric fake quantization. K8V4 and K4V8 have identical average precision and save the same bytes, yet K8V4 wins in five of six families and ties OLMo within 0.002. Across all contexts and held-out distributions it wins 19 of 20 matched comparisons. The effect is largest for Qwen2.5-3B/1.5B: acceptance improves from 0.282 to 0.594 at 1K and from 0.300 to 0.636 at 4K. K8V4 saves 61.7% of the draft cache and 27.0% of combined target-plus-draft KV memory for this pair. Depending on model-size ratio, total savings range from 10.5% (SmolLM2) to 28.8% (Llama).

The asymmetry persists outside the main setting. On two shuffled C4 seeds, K8V4–K4V8 differences are +0.185, +0.282 for Qwen, +0.010, +0.074 for OLMo, and +0.026, +0.090 for Llama. For Qwen at 8K and 16K, the differences remain +0.260 and +0.140, respectively, although these exploratory long-context runs contain only four and two prompts.

5.3 THE ASYMMETRY TRANSFERS TO MATHEMATICS, CODE, AND LONG PREFIXES

At the six-bit all-layer budget, K-priority remains close to native while V-priority fails across held-out domains. On GSM8K answers, K8V4 exceeds K4V8 by 22.72 points at 512-token context (95% CI [19.34, 25.90]) and 25.18 points at 1K (CI [20.76, 29.81]); K8V4 itself is within 0.12 and 0.09

Table 2: Equal-memory K8V4 versus K4V8 at 1K WikiText. Memory includes 16-bit per-vector scales.

Target / draft	K8 V4	K4 V8	Δ accept.	Total KV saved
Qwen2.5 3B/1.5B	0.594	0.282	+0.312	27.0%
Qwen2.5 7B/3B	0.551	0.426	+0.125	24.2%
Qwen3 8B/4B	0.651	0.542	+0.109	23.6%
Llama 3.1 8B/3.2 3B	0.523	0.512	+0.011	28.8%
OLMo-2 7B/1B	0.428	0.430	-0.002	12.3%
SmolLM2 1.7B/360M	0.487	0.390	+0.097	10.5%

points of native acceptance. On HumanEval prompts, the corresponding K8V4–K4V8 gaps are 27.03 points at 256 tokens (CI [21.22, 33.26]) and 32.93 points at 512 (CI [27.32, 38.48]). The HumanEval aggregate contains no non-tie target disagreements.

The separation grows with long PG19 prefixes. At 16K, K-priority is 1.03 points below native (CI [−4.47, 2.14]), while V-priority is 44.76 points below native (CI [−52.50, −38.14]), giving a paired K-priority advantage of 43.73 points (CI [37.07, 51.28]). At 32K, K-priority is 3.37 points below native (CI [−6.89, −0.17]), while V-priority drops by 55.51 points; their paired difference is 52.13 points (CI [43.22, 61.73]). Both contexts use two seeds and contain no non-tie target disagreements. These results show that the naive key quantizer degrades with longer prefixes; they do not distinguish intrinsic routing sensitivity from poor handling of persistent key-channel outliers.

5.4 OUTLIER-AWARE KEY QUANTIZATION REMOVES THE APPARENT FRAGILITY

We next change only the key quantizer geometry: keys use affine scales and offsets per channel over 32-token groups, while values retain the same per-token symmetric quantizer as the diagnostic baseline. Across three 1K Qwen seeds (190 valid paired prompts), K4V8 with a 128-token floating-point residual window changes acceptance by +0.14 points (95% paired bootstrap CI [−0.59, 0.86]) while saving 23.7% of total target-plus-draft KV. Relative to the original per-token K4V8 experiment, changing the key axis recovers 28.70 acceptance points (CI [26.12, 31.31]). K4V4 changes acceptance by only −0.28 points (CI [−1.30, 0.73]) while saving 29.2% total KV. Removing the residual window increases savings to 25.6% for K4V8 and 31.0% for K4V4, with acceptance changes of −0.41 (CI [−1.24, 0.40]) and −1.05 points (CI [−2.01, −0.10]), respectively. These results invalidate an interpretation based on bit-width alone. The three-seed 4K replication agrees: residual-window K4V8 changes acceptance by +0.77 points (CI [−0.57, 2.18]) while saving 25.3% total KV, and its axis gain over per-token K4V8 is 30.01 points (CI [26.46, 33.60]). K4V4 changes acceptance by −0.61 points (CI [−2.43, 1.24]) while saving 30.7% total KV. With no residual window, K4V8 and K4V4 save 25.7% and 31.2% total KV, respectively, with confidence intervals that still include no acceptance change.

A controlled factorial cell keeps affine value quantization fixed, removes the residual window, and changes only the key axis. With per-token keys, equal-memory K8V4 beats K4V8 by 28.70 acceptance points across 186 paired prompts (CI [26.10, 31.38]); with grouped per-channel keys, the contrast becomes −0.33 points across 189 prompts (CI [−1.26, 0.62]). The more aggressive K4V2–K2V4 comparison reverses sign from +15.05 points (CI [13.58, 16.65]) to −3.66 points (CI [−5.80, −1.56]). Teacher-forced quality exhibits the same reversal: the K8V4–K4V8 KL contrast is −2.023 under per-token keys (CI [−2.119, −1.927]), but +0.00440 under per-channel keys (CI [0.00391, 0.00494]). Thus, the key axis alone can reverse both the speculative and ordinary-quality preference while value geometry, model, prompts, and nominal bit budget remain fixed.

The prompt-paired key-group sweep further localizes this dependence. We vary the grouped per-channel key quantizer over 16, 32, 64, and 128 channels with three disjoint seeds and 96 paired prompts per group size; every trajectory matches the independent BF16 target. The mild K8V4–K4V8 contrast is unresolved at every group size. At the aggressive budget, however, K4V2–K2V4 is −5.83 acceptance points for groups of 16 (95% CI [−8.49, −3.31]), −2.94 for 32 (CI [−6.07, +0.09]),

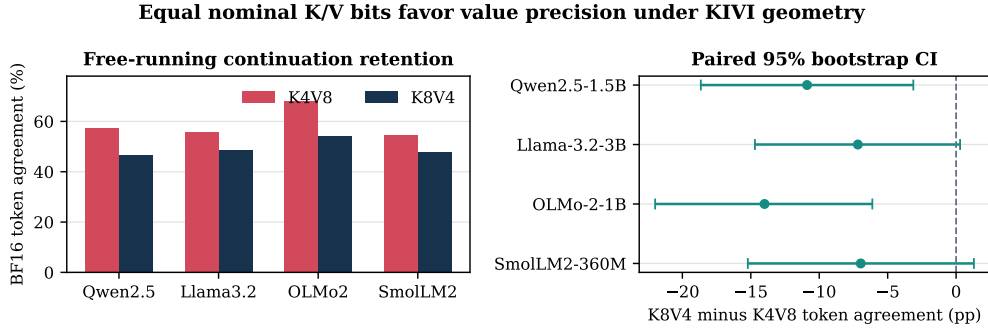


Figure 3: Free-running generation under an equal nominal 12-bit K/V budget. Bars show agreement with each model’s BF16 continuation; the forest plot shows paired K8V4-minus-K4V8 token-agreement differences over 96 prompts per model. K4V8 has the higher point estimate in all four families, with CIs excluding zero for Qwen and OLMo.

−1.57 for 64 (CI [−4.69, +1.61]), and −1.87 for 128 (CI [−4.99, +1.13]). The resolved value-precision preference at fine grouping weakens as key groups grow, so allocation and quantizer granularity must be selected jointly rather than as independent hyperparameters.

A matched sweep over BF16 recent-key residual windows of 0, 32, 128, and 256 tokens shows that the residual is another interacting design choice rather than a universally beneficial safeguard. For K4V4, removing the residual saves 70.56% of draft-cache bytes and 30.87% of combined target-draft KV, versus 66.33% and 29.02% with 128 residual tokens; its acceptance changes by only −0.37 points relative to the 128-token setting (CI [−1.82, +1.05]). A 256-token residual reduces draft savings to 62.10% without a resolved acceptance gain. Allocation contrasts nevertheless interact with the window: K8V4−K4V8 is +1.29 points at 32 tokens (CI [+0.03, +2.68]), whereas K4V2−K2V4 is −3.33 points at 256 tokens (CI [−6.26, −0.42]); the other six contrasts are unresolved. All 12 runs pass prompt-pairing, completeness, and exact-target gates.

We next remove speculative verification from the measurement entirely and run 64-token continuations from 1K-token prefixes under KIVI geometry. Each model uses three disjoint seeds and 96 paired prompts. Figure 3 shows that K4V8 has higher token agreement with BF16 than equal-nominal-budget K8V4 for all four model families. The paired K8V4−K4V8 differences are −10.89 points for Qwen (95% CI [−18.65, −3.12]), −7.18 for Llama (CI [−14.70, +0.29]), −14.00 for OLMo (CI [−21.99, −6.12]), and −6.97 for SmolLM (CI [−15.22, +1.30]). Exact-sequence retention also significantly favors K4V8 for Qwen, OLMo, and SmolLM. This verifier-free result supports a geometry-conditional value-precision preference in ordinary generation; it does not by itself establish the speculative-acceptance ranking.

A longer aggressive control repeats 256-token generation on Llama, OLMo, and SmolLM with three seeds and 144 paired prompts. K4V2 retains 2.74 percentage points less of the BF16-matching prefix than K2V4 (95% hierarchical-bootstrap CI [−6.02, −0.94]). Its overall token-agreement difference is −1.71 points (CI [−4.47, +0.49]), which is directionally consistent but unresolved. K4V4 saves 66.56% of standalone-cache bytes while retaining 24.67% exact token agreement over the full 256-token continuation. This strict trajectory metric is intentionally harsher than downstream task accuracy because one early token change alters every subsequent cache state.

Downstream multiple-choice accuracy is substantially less sensitive than exact free-running trajectory agreement. A strict cross-model aggregate covers Qwen2.5-1.5B, Llama-3.2-3B, OLMo2-1B, and SmolLM2-360M with three disjoint shards per task and no missing or underfilled runs. No K8V4−K4V8 contrast excludes zero. On ARC-Challenge the differences are +0.34, −0.67, 0.00, and −0.34 points for Qwen, Llama, OLMo, and SmolLM, respectively; on HellaSwag they are −0.26, −0.35, +0.17, and +0.52 points. For example, SmolLM’s 95% CIs are [−1.01, 0.00] on 297 ARC examples and [−0.17, +1.39] on 576 HellaSwag examples. K4V4 saves roughly 54–59% of standalone cache bytes while remaining within paired uncertainty of BF16 across these task controls. Exact token retention detects trajectory drift, but it cannot replace task-level quality evaluation.

A confirmatory sequential-verifier study covers six target–draft pairs from Qwen2.5, Qwen3, Llama 3, OLMo 2, and SmolLM2. Across 18 runs and 576 paired prompts, every quantized trajectory matches an independent tokenwise BF16 target. At the mild budget, K4V4 saves 66.13% of draft-cache bytes and 22.64% of combined target–draft KV, with a model-weighted acceptance change of -0.79 points (95% hierarchical-bootstrap CI $[-1.63, 0.04]$). The equal-mean-bit K8V4–K4V8 contrast is -0.05 points (CI $[-0.76, 0.63]$), so mild K/V allocation is unresolved. At the aggressive budget, the K4V2–K2V4 contrast is -1.94 points (CI $[-3.50, -0.40]$), significantly favoring preservation of value precision under grouped per-channel key quantization. This cross-family result and the controlled axis reversal above jointly rule out a universal bit-width-only K/V sensitivity claim.

A fixed-key value-precision sweep identifies a sharp knee below four bits. Across three 1K-context seeds, K8V4 saves 27.00% of total target-plus-draft KV memory with an acceptance change of -1.06 points (CI $[-2.33, 0.19]$). K8V3 increases savings to 28.37% but incurs a small significant drop of 1.45 points (CI $[-2.89, -0.01]$), while K8V2 collapses by 23.74 points. We therefore use K8V4 as the conservative operating point and treat K8V3 as an aggressive option. The same knee holds across three 4K-context seeds: K8V4 changes acceptance by -0.34 points (CI $[-1.95, 1.21]$), K8V3 drops 2.28 points (CI $[-4.25, -0.35]$), and K8V2 drops 24.09 points.

The aggressive asymmetry persists when the speculative proposal length changes. For each $\gamma \in \{2, 4, 8\}$, three disjoint seeds provide 144 paired prompts, and every quantized trajectory matches independent BF16 target-greedy output. The K4V2–K2V4 acceptance contrasts are -5.04 points at $\gamma = 2$ (CI $[-7.60, -2.50]$), -4.80 at $\gamma = 4$ (CI $[-7.30, -2.35]$), and -3.55 at $\gamma = 8$ (CI $[-5.92, -1.29]$). Both policies use the same nominal mean bit-width; because key metadata is asymmetric, their total-cache savings are near rather than exactly equal (31.76% versus 31.38%). The mild K8V4–K4V8 contrast is unresolved at $\gamma = 2$ and 4, but becomes $+1.42$ points at $\gamma = 8$ (CI $[0.37, 2.51]$). Thus, proposal length changes effect size and can resolve a mild preference, while the aggressive value-precision preference remains stable across all three proposal lengths under the deployed KIVI geometry.

The same KIVI geometry remains viable at long prefix lengths. Our strict PG19 aggregate contains six full runs, three seeds per context, 60 prompts, and 360 quantized trajectories that all match independent BF16 target-greedy output. At 16K, K4V4 saves 70.83% of draft-cache bytes and 30.99% of combined target–draft KV, with a -0.94 -point acceptance change across 36 paired prompts (CI $[-2.06, -0.13]$). At 32K, it saves 70.96% of draft KV and 31.04% combined KV, with a -0.73 -point change across 24 prompts (CI $[-2.99, 1.28]$). The mild K8V4–K4V8 contrast is unresolved at 16K ($+0.13$ points, CI $[-1.09, 1.26]$) but favors preserving values at 32K (-3.50 points, CI $[-6.62, -0.45]$). The aggressive K4V2–K2V4 contrast is unresolved at both contexts. A matched teacher-forced PG19 sweep under the same windows, seeds, cache geometry, and strict completeness gate is now complete and shows no statistically resolved objective-preference reversal. Ordinary-quality KL strongly favors K4V8 over K8V4 at both 16K (contrast $+0.00305$, CI $[+0.00210, +0.00427]$) and 32K ($+0.00309$, CI $[+0.00203, +0.00429]$). Acceptance agrees at 32K and is unresolved at 16K. At the aggressive budget, quality strongly favors K2V4 over K4V2 at 16K (KL contrast $+0.0752$, CI $[+0.0531, +0.1039]$) and 32K ($+0.1063$, CI $[+0.0633, +0.1677]$), while acceptance is unresolved at both contexts. Across the five compressed policies, Spearman correlation between acceptance harm and quality KL is 0.60 at 16K and 0.80 at 32K. Teacher-forced KL is therefore a useful screening objective in this setting, though acceptance remains the final serving metric.

A complementary confusable 16-way associative-passkey control reveals that the preferred asymmetry can change again for retrieval. For Qwen2.5-1.5B, the strict aggregate contains all nine 8K, 16K, and 32K model-seed runs, 48 paired examples per context, and no missing or underfilled runs. K4V2 exceeds equal-memory K2V4 by 10.42 accuracy points at 8K (95% CI $[0.00, 20.83]$), 20.83 points at 16K (CI $[8.33, 33.33]$), and 12.50 points at 32K (CI $[2.08, 25.00]$). At 16K, BF16 and K4V2 each retrieve 47/48 passkeys, whereas K2V4 retrieves 37/48; at 32K the three scores are 44/48, 47/48, and 41/48, respectively. Both asymmetric policies save approximately 77% of the standalone cache. A powered 16K replication on 192 disjoint examples confirms the Qwen effect: K4V2 retrieves 189/192 (98.44%) versus 158/192 (82.29%) for K2V4, a $+16.15$ -point paired difference (95% CI $[+10.42, +21.88]$). BF16 retrieves 186/192, while K4V2 and K2V4 save 77.06% and 77.01% of standalone KV, respectively. A strict Llama-3.2-3B replication reverses the resolved 32K preference: K4V2 trails K2V4 by 10.42 points (CI $[-18.75, -2.08]$), scoring 42/48 versus 47/48; its 8K and 16K contrasts remain unresolved. A powered Llama 16K replication remains unresolved: K4V2

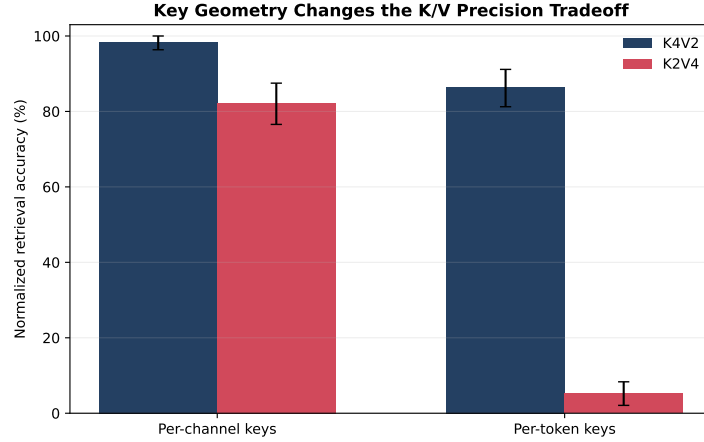


Figure 4: Qwen2.5-1.5B associative-passkey retrieval at 16K over 192 paired examples. Per-token key quantization sharply magnifies the cost of reducing key precision relative to per-channel KIVI-style keys. Error bars are paired bootstrap 95% confidence intervals.

scores 185/192 versus 183/192 for K2V4, a +1.04-point difference (CI $[-1.56, +4.17]$). Qwen3-4B is saturated at 48/48 for every policy at 8K and 16K; at 32K, K4V2 scores 47/48 and K2V4 scores 48/48, an unresolved -2.08 -point contrast (CI $[-6.25, 0.00]$). A powered Qwen3 16K replication is fully saturated: BF16 retrieves 191/192 passkeys and every compressed policy, including K4V2 and K2V4, retrieves 192/192, yielding an exact zero paired contrast. Across the three powered 16K model studies, the macro K4V2–K2V4 effect is $+5.73$ points, but its model-bootstrap interval $[0.00, +16.15]$ includes a null cross-family effect and is driven primarily by Qwen2.5. A strict model-level bootstrap over the smaller three-context study of Qwen2.5, Llama 3, and Qwen3 gives macro K4V2–K2V4 effects of $+4.86$ points at 8K (CI $[0.00, +10.42]$), $+6.25$ at 16K (CI $[-2.08, +20.83]$), and exactly 0.00 at 32K (CI $[-10.42, +12.50]$). These draft-only synthetic results measure retrieval quality rather than speculative acceptance. Together they show that even within one task and quantizer geometry, K/V precision preference is model-dependent rather than universally K-first or V-first.

Quantizer geometry amplifies this asymmetry (Figure 4). On the same 192 Qwen2.5 16K examples, per-channel keys give a K4V2–K2V4 contrast of $+16.15$ points (CI $[+10.94, +21.88]$), whereas per-token keys give $+81.25$ points (CI $[+75.00, +86.98]$). The resolved axis-by-allocation interaction is -65.10 points for per-channel minus per-token contrasts (CI $[-73.44, -56.77]$). Per-token K2V4 retrieves only 10/192 examples, versus 166/192 for per-token K4V2. Thus key fragility is not merely a consequence of using fewer bits: preserving the channel-wise outlier structure identified by KIVI is essential, and bit allocation must be specified jointly with quantizer geometry.

A natural LongBench passage-retrieval control on larger instruction models provides a less synthetic check. Across Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Qwen3-8B (72 paired examples per model), K4V2 matches the BF16 macro accuracy while saving 76.96% of standalone KV. K2V4 saves the same bytes within 0.07 points but changes macro accuracy by -3.24 points (model-bootstrap CI $[-9.72, 0.00]$). The K4V2–K2V4 contrast is $+3.24$ points (CI $[0.00, +10.19]$): Qwen2.5 contributes a resolved $+9.72$ -point effect, whereas Llama and Qwen3 saturate. This supports a geometry-dependent retrieval risk but not a universal cross-family allocation rule.

5.5 ACTUAL PACKED STORAGE REALIZES THE ESTIMATED SAVINGS

Figure 5 replaces the analytical payload with real bit-packed `uint8` tensors, BF16 quantizer metadata, and a BF16 residual key window for a Qwen2.5-1.5B-shaped cache. K4V4 saves 66.80% at 1K, 70.02% at 4K, 70.83% at 16K, and 70.96% at 32K. At 32K this reduces persistent KV storage from 896.0 to 260.2 MiB, closely matching the estimator used throughout the quality campaign.

We additionally implement a direct packed Triton attention microkernel with split-sequence online softmax. It reads packed K/V and quantizer metadata without materializing a context-length BF16 cache. At 32K, its K4V4 output matches the materialized reference with cosine 0.999994 and relative

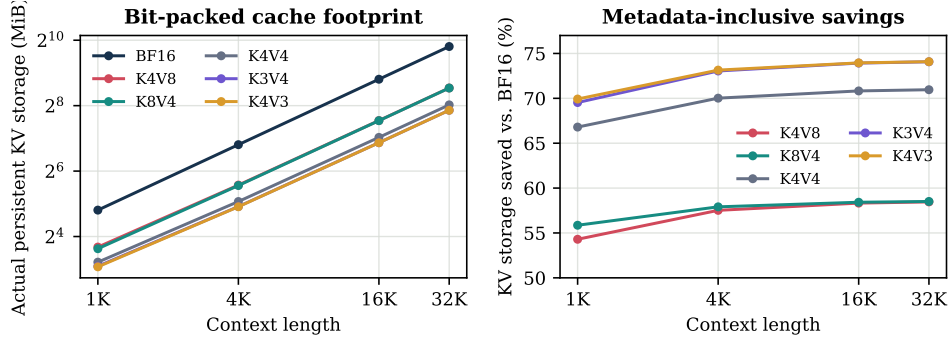


Figure 5: Actual metadata-inclusive packed storage for a 28-layer, two-KV-head, 128-head-dimension cache.

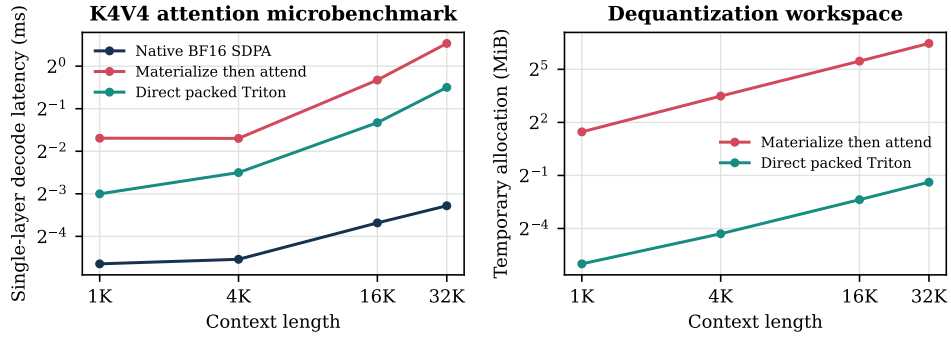


Figure 6: Single-token, single-layer Qwen2.5-1.5B-shaped K4V4 microbenchmark. The direct packed Triton path improves over materialize-then-attend and nearly eliminates temporary cache allocation, but does not yet match native BF16 SDPA. All latency numbers are diagnostic rather than production throughput.

RMSE 0.00342. It takes 0.708 ms per layer, $2.05\times$ faster than the 1.448 ms materialized path, and reduces temporary allocation from 88.3 MiB to 0.38 MiB. Native BF16 SDPA still takes only 0.103 ms, so the direct kernel remains $6.88\times$ slower. The result demonstrates that the compressed representation can be consumed directly and eliminates dequantization workspace, but further kernel engineering or a production backend is required for a throughput gain.

We test whether this conclusion survives model shape, batching, and context. A complete matrix covers six draft shapes (Qwen2.5-1.5B/3B, Qwen3-4B, Llama-3.2-3B, OLMo2-1B, and SmoILM2-360M), batches 1/4/16, contexts 1K/4K/8K/16K/32K, and K4V4, K4V8, and K8V4. Across 18 source runs and 1,512 kernel measurements, K4V4 saves 69.26% of cache bytes on average and is $3.43\times$ faster than materialize-then-attend, but reaches only $0.223\times$ native BF16 speed geometrically. K4V8 and K8V4 save 56.76% and 57.35%, run $3.90\times$ and $3.61\times$ faster than materialization, and reach $0.315\times$ and $0.294\times$ native speed. No matrix cell beats native BF16 SDPA. The worst packed-versus-materialized maximum absolute error is 0.00391, so numerical reconstruction is not the dominant systems bottleneck.

5.6 A CONSERVATIVE LEARNED ALLOCATION

Profiling the top eight draft layers and selecting precision under a five-point additive acceptance budget gives deliberately conservative allocations. Across the six pairs, held-out acceptance retention ranges from 94.9% to 102.8%, while total KV savings range from 2.9% to 9.9%. Figure 7 shows that uniform K4V4 often saves more memory but can retain less than half of native acceptance; the sensitivity-aware solutions stay near the 100% line.

A held-out retrieval-aware successor evaluates disjoint associative-passkey indices 288–383 on Qwen2.5-1.5B, Llama-3.2-3B, and Qwen3-4B. It does not improve over the strongest uniform policy:

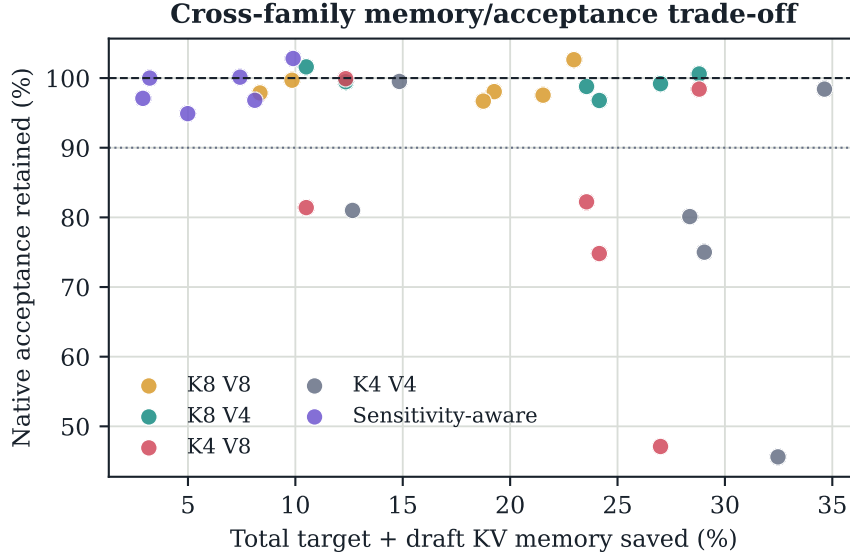


Figure 7: Memory–acceptance trade-off at 1K. Sensitivity-aware allocations are conservative; K8V4 provides a stronger uniform operating point than K4V8.

on Qwen, the learned mixed policy and uniform K4V2 both score 89/96, while uniform K2V4 scores 92/96; Llama and Qwen3 are saturated across policies. We therefore report the learned allocator as a null result rather than selecting it post hoc.

5.7 MATCHED DOWNSTREAM OBJECTIVES MOSTLY AGREE

We directly evaluate the same 20 non-native K/V configurations under speculative acceptance and teacher-forced ordinary-LM quality. At 1K, K4V4 saves 66.3% of the draft cache and 29.0% of total target-plus-draft KV; its acceptance change is -0.43 points (95% CI $[-1.41, 0.55]$), with KL 0.00781 (CI $[0.00718, 0.00851]$). At 4K, residual overhead is amortized: K4V4 saves 69.9% of draft KV and 30.6% total, with acceptance change -0.05 points (CI $[-1.76, 1.73]$) and KL 0.00837 (CI $[0.00728, 0.00963]$). It is the maximum savings configuration satisfying our conservative two-point acceptance and 0.01-KL budgets at both contexts.

The objectives are not independent. Acceptance harm and quality KL have Spearman rank correlation 0.964 at 1K and 0.878 at 4K. Across ten predeclared paired comparisons, including six within 0.2 percentage points of total-cache savings, we observe no preference reversal. In particular, matched K4V3 versus K3V4 comparisons favor K3V4 under both objectives, indicating that values require more precision under this outlier-aware key quantizer. Objective-specific choices can differ at coarse minimum-savings thresholds, but their measured regrets are small and do not establish a distinct speculative allocation rule.

A powered Qwen2.5-7B/3B FineWeb-Edu replication confirms this result on three disjoint streaming shards. Across 1,526 valid paired speculative prompts, K4V3-minus-K3V4 acceptance is -0.65 points (95% CI $[-1.16, -0.15]$). Relative to native BF16 draft caches, K3V4 changes acceptance by only -0.06 points (CI $[-0.46, +0.34]$) while saving 69.03% of draft-cache bytes and 27.01% of combined target-plus-draft KV. Across 768 teacher-forced sequences, K4V3-minus-K3V4 KL is $+0.00848$ (CI $[+0.00777, +0.00927]$), again favoring K3V4. The powered test therefore rejects both the exploratory objective reversal and a geometry-independent rule that keys inherently require more precision than values.

An all-layer SmolLM2-360M replication provides a direct aggressive-budget stress test. Across 288 prompts and 2,016 exact target trajectories at 1K and 4K, the equal-memory K4V2–K2V4 acceptance contrast is unresolved: -1.34 points at 1K (CI $[-3.24, +0.54]$) and $+0.63$ points at 4K (CI $[-1.15, +2.49]$). Ordinary-quality KL nevertheless favors K2V4 at both contexts by 0.03128

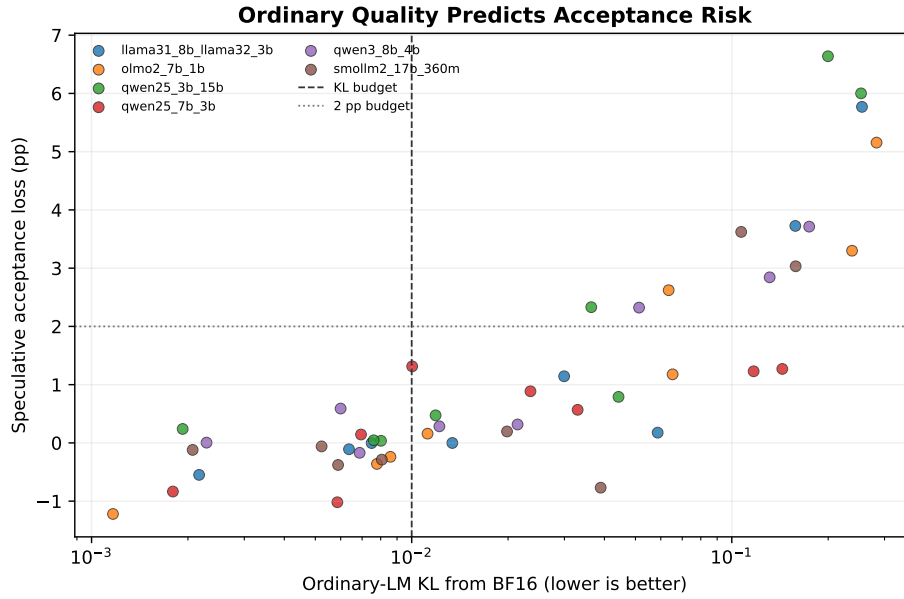


Figure 8: Ordinary-LM quality is a useful but incomplete surrogate for speculative acceptance harm across six held-out model pairs. The dashed line denotes the conservative $\text{KL} \leq 0.01$ screen.

(CI [+0.02541, +0.03767]) and 0.04353 (CI [+0.03154, +0.05935]), respectively. The 4K means reverse direction, but the acceptance interval rules out a resolved objective-specific allocation.

The corresponding all-layer Qwen2.5-3B/1.5B replication is complete across 1K and 4K contexts with three seeds and 288 prompts. At 1K, K4V2 trails K2V4 by 2.72 acceptance points (95% CI [-4.76, -0.66]); at 4K the mean gap is -2.59 points but remains unresolved (CI [-5.88, +0.70]). Ordinary-quality KL strongly favors K2V4 at both contexts: K4V2-K2V4 is +0.12186 at 1K (CI [+0.10334, +0.14284]) and +0.12675 at 4K (CI [+0.08585, +0.18014]). The policies use the same mean bit budget and have near-matched measured storage: their total-cache savings differ by 0.38 points at 1K and 0.10 points at 4K. All target trajectories pass the strict sequential exactness gate. Thus the powered Qwen control shows agreement, not reversal: both downstream objectives favor retaining more value precision under the tested outlier-aware key quantizer.

OLMo2-7B/1B independently reproduces this aggressive all-layer result. Across the same 1K/4K, three-seed design, all six speculative and six quality runs are complete and all 2,016 target trajectories are exact. K4V2 trails near-matched K2V4 by 1.91 acceptance points at 1K (95% CI [-3.70, -0.09]) and 2.35 points at 4K (CI [-4.58, -0.07]). Ordinary-quality KL agrees: K4V2-K2V4 is +0.05148 at 1K (CI [+0.03861, +0.06648]) and +0.04786 at 4K (CI [+0.03941, +0.05964]). Total-cache savings differ by only 0.17 and 0.04 percentage points, respectively. At the mild K8V4-K4V8 comparison, acceptance remains unresolved while KL favors K4V8. Together, Qwen and OLMo provide two powered families where aggressive value precision improves both speculative acceptance and ordinary quality.

This agreement is useful operationally. Across 48 model-pair/configuration cells, ordinary-quality KL and speculative-acceptance harm have Spearman correlation 0.849; the macro average across the six model pairs is 0.794 (95% model-pair bootstrap CI [0.631, 0.929]). A leave-one-model-pair-out linear predictor using only log-KL and top-1 agreement predicts acceptance harm with 1.01-point RMSE, 0.70-point MAE, and $R^2 = 0.724$. A conservative KL threshold of 0.01 selects 19 of 48 cells, all within a two-point acceptance-loss budget, but recovers only 54.3% of all cells satisfying that budget. Ordinary quality can therefore act as a cheap conservative screen, while direct acceptance evaluation remains useful for recovering more aggressive candidates. These empirical rates are descriptive of the tested cells rather than a distribution-free safety guarantee.

Larger Target Models Are More Robust to the Same KV Quantization

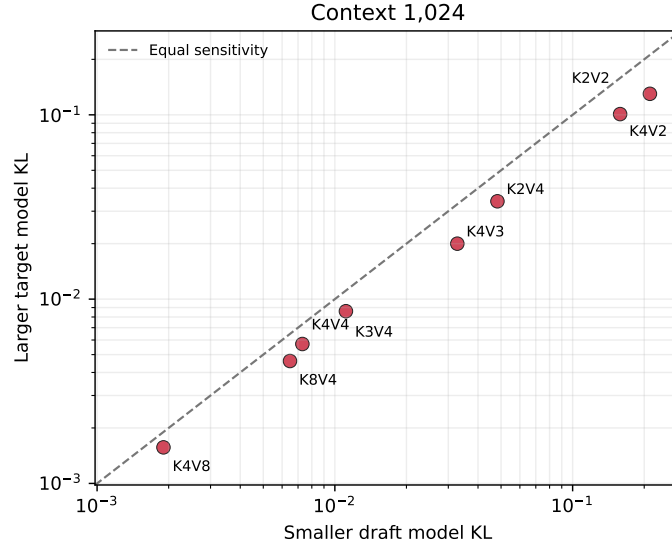


Figure 9: Ordinary-LM KL under identical KV precision for the larger target and smaller draft checkpoints, macro-averaged across six model families. Every setting lies below the equal-sensitivity diagonal.

5.8 LARGER TARGET CHECKPOINTS ARE MORE QUANTIZATION-ROBUST

Ordinary-decoding robustness also depends on which model owns the cache. We apply the same eight K/V precision settings separately to the larger target and smaller draft checkpoint in each of the six model pairs. The target has lower KL in all 48 paired model-configuration comparisons. At K4V4, target KL is 0.00571 versus 0.00730 for the draft (paired macro difference -0.00159 , 95% model-pair bootstrap CI $[-0.00276, -0.00068]$); at K4V8, it is 0.00157 versus 0.00190 (difference -0.00033 , CI $[-0.00046, -0.00021]$). Figure 9 shows the effect across the full precision grid.

Because target role is confounded with model size in standard speculative pairs, we interpret this as evidence for role-and-scale-aware calibration rather than a causal target-role effect. Moreover, a quantized verifier is no longer exact: lower teacher-forced KL does not eliminate rare top-1 changes. Draft-only compression therefore remains the lossless operating mode, while target-cache compression requires explicit quality and exactness budgets.

5.9 JOINT COMPRESSION EXPOSES EXACT AND APPROXIMATE REGIMES

We evaluate 25 target/draft precision combinations at 1K and 4K, then maximize total KV savings subject to target KL, target top-1 agreement, runtime fidelity, and a two-point lower-confidence acceptance budget. At 1K, the selected joint policy is target K4V8 with draft K8V4. It saves 54.59% of total KV, changes acceptance by -0.46 points (95% CI $[-1.78, 0.84]$), and gives target KL 0.00177 (upper CI 0.00190) with 97.69% target top-1 agreement. This is an approximate operating point: five of 288 prompt occurrences are classified as non-tie or unknown target disagreements, versus two for the BF16 baseline.

Keeping the target cache unquantized gives a distribution-preserving alternative. Draft K4V4 saves 29.02% of total KV at 1K with a $+0.16$ -point acceptance change (CI $[-0.90, 1.25]$), and 30.58% at 4K with a $+0.39$ -point change (CI $[-1.13, 1.89]$). At 4K, no target-quantized candidate satisfies the predeclared two-point acceptance lower-confidence constraint, so the selector returns this exact-target policy. Figure 10 visualizes both choices. Joint compression therefore offers substantially greater memory reduction, but not a uniform lossless upgrade over draft-only compression.

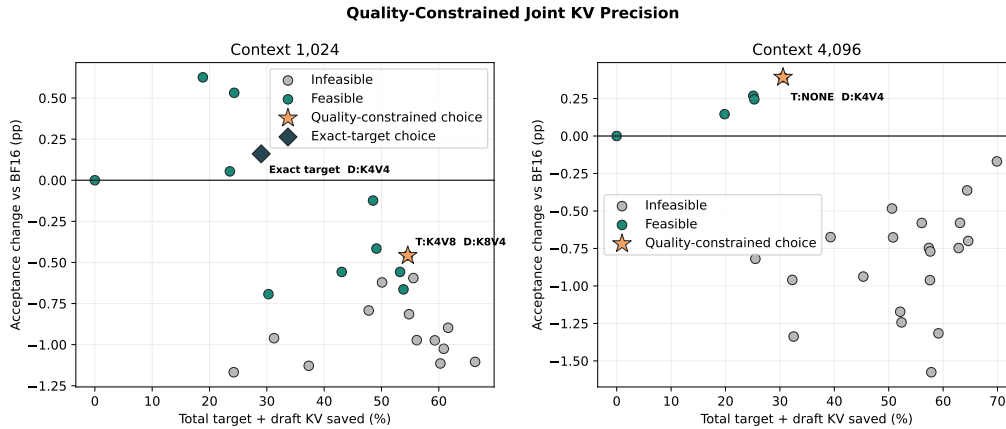


Figure 10: Quality-constrained joint target/draft KV precision. Stars mark the maximum-savings feasible policy; the diamond marks the best policy that keeps the target cache unquantized. At 4K these choices coincide.

5.10 OBJECTIVE-SPECIFIC PROFILES DO NOT IMPROVE ACCEPTANCE CONSISTENTLY

Optimizing the sensitivity profile under the speculative loop is not reliably better than profiling ordinary language-model quality. In a powered Qwen replication with 384 held-out prompts, the acceptance-mass allocation changes several layer-wise K/V assignments but trails the quality allocation by 0.94 percentage points (95% CI $[-1.76, -0.14]$). The corresponding effects are -1.74 points for Llama (CI $[-4.47, 0.86]$) and -0.10 points for OLMo (CI $[-2.49, 2.33]$). Under the more aggressive three-bit mean budget on Qwen, acceptance optimization is significantly worse by 4.93 points (CI $[-7.29, -2.64]$).

These results reject the strongest version of our initial hypothesis: directly optimizing a small calibration set for speculative acceptance is not sufficient to discover a better held-out allocation. Greedy acceptance is discrete and high variance, while the additive allocator ignores interactions among jointly quantized components. In contrast, quality profiling repeatedly recovers the stable K-first solution and has lower held-out KL. We therefore treat downstream acceptance as the evaluation criterion, but not yet as a proven superior training objective.

A stricter metadata-aware follow-up reaches the same conclusion under exactly matched profiled bytes. It evaluates 48 held-out run cells at 1K and 4K, with three seeds per context, and audits 3,840 target trajectories without a mismatch. At the six-bit budget, quality and acceptance profiling independently select the same K4V8 policy in all eight profiled layers; at eight bits all selectors reduce to K8V8. At the aggressive three-bit budget, quality selects K2V4 uniformly, whereas acceptance profiling selects heterogeneous two-, four-, and eight-bit value precisions at the same 6,110,208 profiled saved bytes. On held-out prompts, the acceptance-profiled policy is 0.61 acceptance points worse than the quality policy (hierarchical 95% CI $[-1.98, +0.60]$) and has KL larger by 0.01485 (CI $[+0.01120, +0.02060]$). At five bits both objectives select K2V8. Thus, whenever the objectives disagree here, the acceptance profile does not deliver a resolved held-out acceptance benefit and substantially worsens ordinary quality.

An OLMo-2-7B/1B replication passes the same 48/48-cell, exact-byte, and 3,840/3,840 target-exactness gates. At the three-bit budget, both objectives select K2V4 in all eight profiled layers; at five bits, both select K2V8. Consequently, the cross-objective acceptance and KL effects are exactly zero at both budgets. Relative to native draft KV, K2V4 changes acceptance by -0.91 points (95% CI $[-2.22, +0.48]$), while K2V8 changes it by -0.88 points (CI $[-2.17, +0.43]$). Together with Qwen, this shows that objective-specific calibration differences are neither reliably beneficial nor consistent across model families.

The matched OLMo mild-budget control is also complete: all 24 held-out cells pass evaluator, byte-matching, and 3,840/3,840 target-exactness gates. Quality and acceptance profiling select the same top-eight K4V8 layout at budget six and the same K8V8 layout at budget eight, so their paired

acceptance, KL, and NLL contrasts are identically zero. Relative to native draft KV, the selected budget-six policy changes acceptance by +0.06 points (hierarchical 95% CI $[-0.32, +0.47]$), while budget eight changes it by +0.11 points (CI $[-0.20, +0.43]$). A cross-context K8V4 heuristic control is 0.41 acceptance points below K4V8 (CI $[-1.01, +0.16]$) and incurs +0.00140 ordinary-quality KL (CI $[+0.00092, +0.00214]$). The mild control therefore preserves behavior but neither separates the downstream objectives nor supports a universal K-first allocation rule.

A matched SmolLM2 mild-budget control reaches the same conclusion. All 24 held-out cells pass byte-matching and 3,840/3,840 target-exactness checks. Quality and acceptance profiling select identical allocations at both six- and eight-bit budgets, making every paired cross-objective acceptance, KL, and NLL contrast exactly zero. This independent null shows that the lack of objective separation at mild precision is not specific to OLMo.

A strict SmolLM2-1.7B/360M replication provides the third aggressive family and passes 3,840/3,840 target-exactness checks. At the three-bit budget, the acceptance-profiled allocation improves acceptance over the quality allocation by only 0.11 points (95% CI $[-0.53, +0.77]$), while increasing quality KL by 0.00275 (CI $[+0.00248, +0.00310]$) and delta NLL by 0.00325 (CI $[+0.00107, +0.00548]$). At five bits, the policies are identical. The three-family macro acceptance effect at budget three is -0.17 points, with a model-bootstrap interval of $[-0.61, +0.11]$. Thus, the additive sensitivity allocator does not establish a downstream-objective benefit.

The final exact-byte meta-analysis extends this comparison to five model pairs: Qwen2.5 3B/1.5B and 7B/3B, Llama 3.1/3.2, OLMo 2, and SmolLM2. All ten mild and aggressive matrices are complete, with 38,400/38,400 exact target checks and no ties or invalid prompts. At the aggressive three-bit budget, acceptance-optimized allocation is 0.48 acceptance points worse than quality-optimized allocation (95% model-bootstrap CI $[-0.98, -0.05]$), while increasing ordinary-quality KL by 0.00561 (CI $[+0.00180, +0.01030]$). At aggressive budget five and mild budgets six and eight, both objectives select identical policies. The completed matrix therefore provides evidence against, rather than for, a separate acceptance-trained additive allocator.

We therefore add a direct all-layer equal-memory control that avoids the additive allocator. In a 32-prompt SmolLM pilot with 224/224 exact target checks, K4V2 exceeds K2V4 by 2.39 speculative-acceptance points, although its CI remains unresolved at $[-1.75, +6.73]$. Ordinary quality shows the opposite, resolved preference: K4V2 minus K2V4 is +0.03721 KL (CI $[+0.02255, +0.05279]$) and +0.05423 delta NLL (CI $[+0.02223, +0.08759]$), where lower is better. Both policies remove about 71% of draft-cache bytes and 12.3% of total target-plus-draft KV. We treat this as a pilot preference reversal and pre-register powered 1K/4K, three-seed replications on SmolLM2, Qwen2.5, and OLMo2 before making a main claim.

5.11 FINITE-PRECISION VERIFICATION AUDIT

In all draft-only experiments, the target cache and weights remain BF16; joint target quantization is reported separately above. Nevertheless, multi-token verification and tokenwise greedy decoding invoke different GPU kernel shapes. Across 2,010 prompt-configuration outputs, 308 sequence disagreements occur where either target path has top-1 margin at most 10^{-3} , and 33 (1.64%) remain above that threshold. Focused diagnostics found no causal-mask leakage and no causal-suffix violation. BF16 SDPA reproduces one speculative mismatch on each of four independent non-tie prompts, whereas FP32 SDPA produces zero on all four. Two BF16 traces also show rollback-path top-1 flips, but these disappear in FP32; on an additional traced prompt, switching BF16 from SDPA to eager attention also eliminates the mismatch. We then run a powered backend-dtype audit on the same 32 prompts, with 512 speculative decisions per condition. BF16 SDPA has six top-1 disagreements (one numerical tie and five above the 10^{-3} target-margin threshold), while BF16 eager has seven (four ties and three above threshold). Both FP32 SDPA and FP32 eager have zero disagreements. This precision dependence supports finite-precision kernel-path drift rather than stale-cache corruption. We consequently do not claim bitwise equality to tokenwise BF16 greedy decoding, while the target verification rule itself remains unquantized.

5.12 MOTIVATING NEGATIVE RESULT: DIRECT CROSS-MODEL CACHE MAPS

Our earlier approach attempted to replace parts of the draft cache with linear maps from target-model KV states. Sharing the top four of 28 draft layers reduced total KV memory by only 6.25% and reduced acceptance from 0.497 to 0.455 in the cached-quality benchmark. Sharing every layer degraded acceptance much more. This failure motivated the current direction: preserve each model’s native cache semantics, compress the cheaper draft state, and optimize the quantity that matters for speculative decoding. Concurrent work on cross-model prefill reuse also finds that transfer quality depends strongly on error placement and often benefits from RoPE removal, cross-layer selection, or nonlinear maps (Heo et al., 2026).

6 RELATED WORK

KIVI establishes that key and value caches have different outlier structures and uses per-channel K and per-token V quantization (Liu et al., 2024). KV-AdaQuant argues for allocating more precision to keys based on norm and quantization sensitivity (Hariri et al., 2025). Our work does not claim that K/V asymmetry is new; it tests and optimizes that asymmetry under speculative acceptance, where a single changed proposal can truncate an entire draft block.

Layer-wise mixed precision is also established prior art. KVTuner searches hardware-friendly K/V precision pairs using attention-output sensitivity (Li et al., 2025), while RateQuant calibrates quantizer-specific rate-distortion curves before allocating bits (Zuo et al., 2026). Block-GTQ further allocates key precision at the RoPE frequency-block level (Liang et al., 2026). Consequently, neither mixed precision nor a generic sensitivity allocator is our novelty. Our controlled variable is the downstream objective: we fit and cross-evaluate allocations under ordinary quality and speculative acceptance using the same model, prompts, quantizer, and memory budget.

KVarN shows that cache-quantization errors accumulate differently during autoregressive generation than in prefill-like evaluation and proposes variance normalization to stabilize long reasoning trajectories (Muller et al., 2026). We therefore do not claim the broader observation that decode-time evaluation matters. Our narrower question is whether the verification dynamics of speculative decoding induce a distinct precision policy, and how that policy interacts with key axis, grouping, zero-points, and the full-precision target safety net.

QuantSpec combines self-speculative decoding with a hierarchical 4-bit KV cache (Tiwari et al., 2025). QSpec combines complementary model quantization schemes and reuses model state between drafting and verification (Zhao et al., 2025). In contrast, we study standard two-model speculative decoding and allocate draft-cache precision by measured acceptance sensitivity.

7 LIMITATIONS AND NEXT EXPERIMENTS

The quality evaluator uses KIVI-style grouped per-channel affine keys, per-token affine values, and an optional residual window through fake quantization. Our Triton microkernel directly consumes actual bit-packed payloads and validates numerical fidelity and workspace reduction, but it is a single-token, single-layer diagnostic without production attention features or kernel autotuning. It is slower than native BF16 SDPA on A5000 GPUs. Therefore packed bytes and transient allocation, not end-to-end throughput, are currently the reliable systems metrics. Although the campaign spans six pairs and includes two C4 seeds for three pairs, prompt counts remain modest at 16K–32K and on task-conditioned code and mathematics prompts. Our LongBench control covers passage retrieval only and saturates for two of three instruction models; broader LongBench tasks and larger PG19, HumanEval, and GSM8K samples are needed. Our BF16 audit also shows that mathematically distribution-preserving speculation need not be bitwise identical to tokenwise decoding across kernel shapes; higher-precision and production-kernel audits remain important.

The additive allocation in Equation 10 ignores interactions between compressed components. We plan to fit a small risk predictor over joint allocations and compare it against uniform precision, KIVI-style quantization, KV-AdaQuant, KVTuner, RateQuant, Block-GTQ, QuantSpec, and random allocations at matched memory. A fused packed-cache kernel in vLLM, SGLang, or a standalone

Triton harness with production-level tiling and scheduling is required before making end-to-end throughput and serving-capacity claims.

8 CONCLUSION

KV compression should be evaluated by the downstream behavior it preserves, not reconstruction error or synthetic perturbation sensitivity alone. Our initial finding that keys tolerate less Gaussian noise is real under that intervention, but it does not imply that keys universally require more bits. Precision allocation cannot be interpreted independently of quantization axis, grouping, and zero-points: a naive quantizer can manufacture an apparent K/V asymmetry that an outlier-aware quantizer removes. Across six model pairs, changing quantizer geometry shifts the K8V4–K4V8 acceptance contrast by -11.12 points. In the controlled same-model axis intervention, changing only the key axis moves the contrast from $+28.70$ to -0.33 points and reverses the two-bit K/V preference. Our final analysis therefore compares speculative acceptance and ordinary language-model quality using the same KIVI-style quantizer at matched memory. Uniform K4V4 removes 66.13% of the draft cache and 22.64% of combined KV with no statistically resolved acceptance loss and KL below 0.01, but matched objective rankings remain strongly correlated and show no statistically resolved equal-memory reversal. Joint quantization can raise total savings to 54.59% under quality and acceptance constraints, but it changes the target distribution and is not consistently feasible at 4K. Powered retrieval experiments expose a large, resolved interaction between key axis and K/V allocation, while a held-out learned layer policy provides no benefit over uniform baselines. The strict objective matrices further show that our acceptance-specific additive objective is unnecessary: it matches quality-based selection at mild budgets and performs worse at the aggressive budget. These findings support task-aware evaluation and quantizer-aware allocation, not a speculative-specific objective or universally K-first policy. The observed cross-family correlation supports a two-stage calibration procedure: ordinary-quality metrics cheaply reject risky configurations, then direct acceptance testing searches the remaining memory frontier. These findings motivate role-and-scale-aware calibration, packed asymmetric kernels, and allocation policies trained against the downstream decoding objective rather than reconstruction error alone.

REFERENCES

- Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling. *arXiv preprint arXiv:2302.01318*, 2023.
- Mohsen Hariri, Lam Nguyen, Sixu Chen, Shaochen Zhong, Qifan Wang, Xia Hu, Xiaotian Han, and Vipin Chaudhary. More for keys, less for values: Adaptive KV cache quantization. *arXiv preprint arXiv:2502.15075*, 2025.
- Taekyung Heo, Rasoul Shafipour, Ritchie Zhao, Maximilian Golub, Mohammad Mahdi Kamani, Ritika Borkar, Makesh Tarun Chandran, Pantea Zardoshti, and Bitu Darvish Rouhani. Cross-model KV cache transfer in LLM families: A closed-form linear mapping for prefill reuse. *arXiv preprint arXiv:2608.03893*, 2026.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *Proceedings of the 40th International Conference on Machine Learning*, pp. 19274–19286, 2023.
- Xing Li, Zeyu Xing, Yiming Li, Linping Qu, Hui-Ling Zhen, Wulong Liu, Yiwu Yao, Sinno Jialin Pan, and Mingxuan Yuan. KVTuner: Sensitivity-aware layer-wise mixed-precision KV cache quantization for efficient and nearly lossless LLM inference. *arXiv preprint arXiv:2502.04420*, 2025.
- Fengfeng Liang, Yuechen Zhang, and Jiaya Jia. RoPE-aware bit allocation for KV-cache quantization. *arXiv preprint arXiv:2606.24033*, 2026.
- Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *Proceedings of the 41st International Conference on Machine Learning*, pp. 32332–32344, 2024.

918 Lorenz K. Muller, Philippe Bich, Chiara Boretti, Hyun-Min Chang, Jiawei Zhuang, and Lukas
919 Cavigelli. KVarN: Variance-normalized KV-cache quantization mitigates error accumulation in
920 reasoning tasks. *arXiv preprint arXiv:2606.03458*, 2026.

921
922 Rishabh Tiwari, Haocheng Xi, Aditya Tomar, Coleman Hooper, Sehoon Kim, Maxwell Horton, Mah-
923 yar Najibi, Michael W. Mahoney, Kurt Keutzer, and Amir Gholami. QuantSpec: Self-speculative
924 decoding with hierarchical quantized KV cache. In *Proceedings of the 42nd International Confer-*
925 *ence on Machine Learning*, pp. 59668–59686, 2025.

926 Juntao Zhao, Wenhao Lu, Sheng Wang, Lingpeng Kong, and Chuan Wu. QSpec: Speculative
927 decoding with complementary quantization schemes. In *Proceedings of the 2025 Conference on*
928 *Empirical Methods in Natural Language Processing*, pp. 4779–4795, 2025.

929 Fei Zuo, Zikang Zhou, Hao Cong, Xiaoyan Xi, and Ho Fai Leung. RateQuant: Optimal mixed-
930 precision KV cache quantization via rate–distortion theory. *arXiv preprint arXiv:2605.06675*,
931 2026.

932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971